KR-ACC-56 Optical Transport Failure Resolution SOP for Arista Access Switches at KR Puram Access

Node: KR-ACC-56 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_4235

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in the Access role is primarily attributed to the degradation of SFP/XFP modules or physical damage to the optical fibers. This can occur due to the exposure of the fiber to harsh environmental conditions such as extreme temperatures, moisture, or physical stress. The Arista Access Switches utilize SFP/XFP modules for optical transport, which are susceptible to degradation over time. Additionally, the optical fibers connecting the switches may be prone to cuts or damage, disrupting the optical signal transmission. In this SOP, we will diagnose and resolve the Optical Transport Failure by isolating the root cause and performing necessary repairs or replacements.

Pre-requisite Checks

- Verify that the KR-ACC-56 node is accessible and can be reached via the network
- Check the status of the redundant links at the KR Puram Access site
- Ensure that the remote hands are dispatched to the site for physical intervention (if necessary)
- Verify that the necessary tools and spare parts are available at the site

Potential Root Causes

Cause	Probability
SFP/XFP module degradation	High
Fiber cuts or damage	Medium
Rx/Tx power loss due to environmental factors	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface statistics, including SFP/XFP status and optical signal levels, Description: This command will provide an overview of the interface status, including SFP/XFP module status and optical signal levels, which can help identify potential issues

Command: show sfp detail, Expected Output: Detailed SFP/XFP module information, including temperature, voltage, and optical signal levels, Description: This command will provide detailed information about the SFP/XFP module, including temperature, voltage, and optical signal levels, which can help identify potential issues

Command: show log | include ALARM_4235, Expected Output: Alarm log entries related to Optical Transport Failure, Description: This
```

```
command will provide log entries related to Optical Transport Failure, which can help identify the cause of the issue
```

Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP module, Command: replace sfp , Description: Replace the faulty SFP/XFP module with a new one to resolve the Optical Transport Failure

Step 2: Step: Repair or replace the damaged fiber, Command: no shutdown interface , Description: Repair or replace the damaged fiber to restore the optical signal transmission

Step 3: Step: Adjust the Rx/Tx power levels, Command: set interface rx-power tx-power , Description: Adjust the Rx/Tx power levels to ensure optimal optical signal transmission

Post-Check Verification

```
Command: show interface status, Expected Output: Interface statistics, including SFP/XFP status and optical signal levels, Description: Verify that the issue is resolved by checking the interface status and optical signal levels

Command: show sfp detail, Expected Output: Detailed SFP/XFP module information, including temperature, voltage, and optical signal levels, Description: Verify that the SFP/XFP module is functioning correctly by checking the detailed information

Command: show log | include ALARM_4235, Expected Output: No alarm log entries related to Optical Transport Failure, Description: Verify that the issue is resolved by checking the log entries for Optical Transport Failure
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace sfp , Description: Restore the original SFP/XFP module if the replacement fails

- Step: Restore the original fiber, Command: shutdown interface , Description: Restore the original fiber if the repair or replacement fails

- Step: Restore the original Rx/Tx power levels, Command: set interface rx-power tx-power , Description: Restore the original Rx/Tx power levels if the adjustment fails